

EmbedUR Networks Training

Assessment - 1

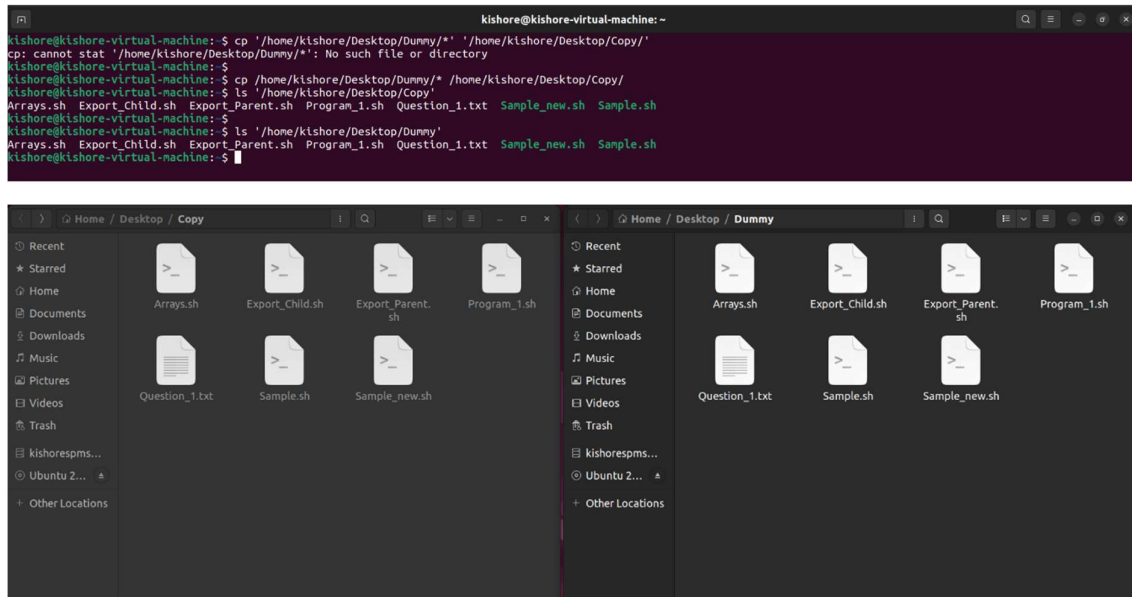
Module - 1: Communication Fundamentals

&

Module - 2 : Networking Architecture and Protocols

Name: Kishore S

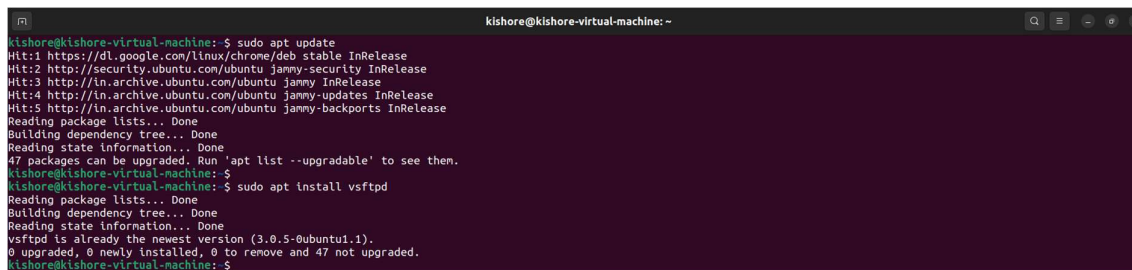
Question – 1: Consider a case, a folder has multiple files and how would copy it to destination machine path (Try using SCP, cp options in Linux)



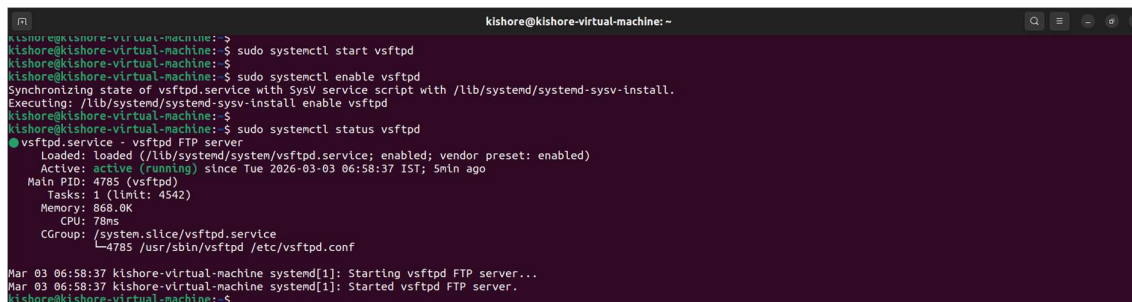
Question – 2: Host a FTP and SFTP server and try PUT and GET operations.

FTP Server

Step - 1: Install the FTP Server

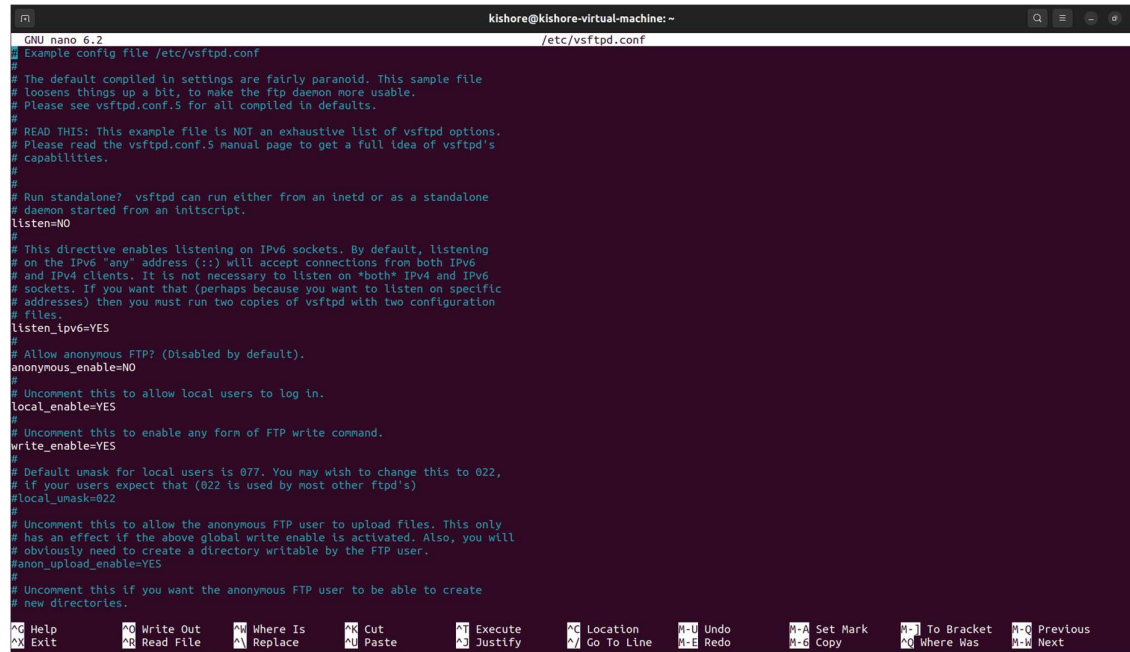


Step - 2: Start and Enable the FTP Server:



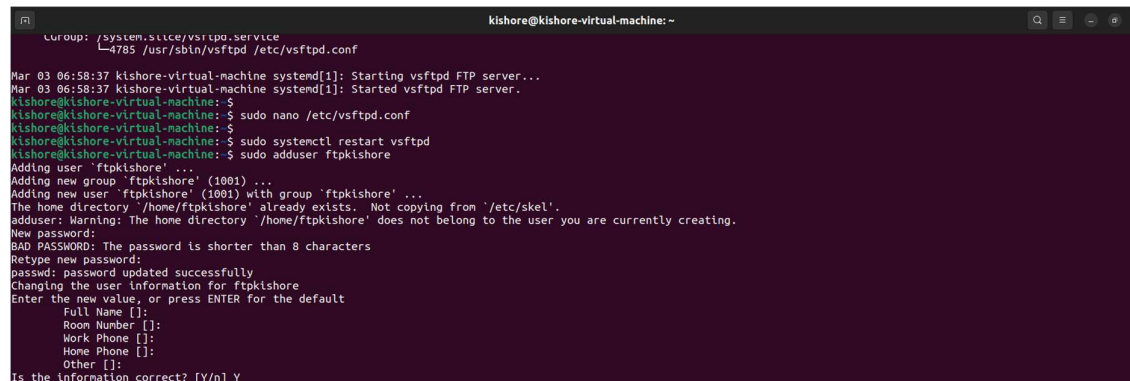
Step - 3: Edit the Config File to Enable Write Mode:

`sudo nano /etc/vsftpd.conf`



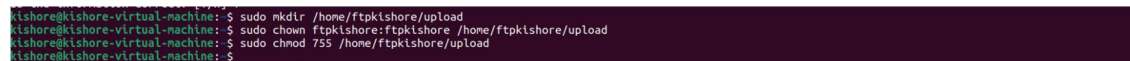
```
kishore@kishore-virtual-machine: ~  
GNU nano 6.2 /etc/vsftpd.conf  
# Example config file /etc/vsftpd.conf  
#  
# The default compiled in settings are fairly paranoid. This sample file  
# loosens things up a bit, to make the ftp daemon more usable.  
# Please see vsftpd.conf.5 for all compiled in defaults.  
#  
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.  
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's  
# capabilities.  
#  
# Run standalone? vsftpd can run either from an inetd or as a standalone  
# daemon started from an initscript.  
listen=NO  
#  
# This directive enables listening on IPv6 sockets. By default, listening  
# on the IPv6 "any" address (:::) will accept connections from both IPv6  
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6  
# sockets. If you want that (perhaps because you want to listen on specific  
# addresses) then you must run two copies of vsftpd with two configuration  
# files.  
listen_ipv6=YES  
#  
# Allow anonymous FTP? (Disabled by default).  
anonymous_enable=NO  
#  
# Uncomment this to allow local users to log in.  
local_enable=YES  
#  
# Uncomment this to enable any form of FTP write command.  
write_enable=YES  
#  
# Default umask for local users is 077. You may wish to change this to 022,  
# if your users expect that (022 is used by most other ftpd's)  
local_umask=022  
#  
# Uncomment this to allow the anonymous FTP user to upload files. This only  
# has an effect if the above global write enable is activated. Also, you will  
# obviously need to create a directory writable by the FTP user.  
anon_upload_enable=YES  
#  
# Uncomment this if you want the anonymous FTP user to be able to create  
# new directories.  
#  
^G Help      ^O Write Out  ^W Where Is   ^X Cut        ^J Execute    ^C Location   ^U Undo       ^M Set Mark   ^I To Bracket  ^Q Previous  
^X Exit      ^R Read File  ^A Replace    ^U Paste      ^_ Justify    ^G Go To Line ^E Redo       ^V Copy       ^N Where Was  ^N Next
```

Step - 4: Restart the Server and Add new User:



```
kishore@kishore-virtual-machine: ~  
Group: /system.slice/vsftpd.service  
└─4785 /usr/sbin/vsftpd /etc/vsftpd.conf  
Mar 03 06:58:37 kishore-virtual-machine systemd[1]: Starting vsftpd FTP server...  
Mar 03 06:58:37 kishore-virtual-machine systemd[1]: Started vsftpd FTP server.  
kishore@kishore-virtual-machine: ~  
kishore@kishore-virtual-machine: ~$ sudo nano /etc/vsftpd.conf  
kishore@kishore-virtual-machine: ~$ sudo systemctl restart vsftpd  
kishore@kishore-virtual-machine: ~$ sudo adduser ftpkishore  
Adding user 'ftpkishore' ...  
Adding new group 'ftpkishore' (1001) ...  
Adding new user 'ftpkishore' (1001) with group 'ftpkishore' ...  
The home directory '/home/ftpkishore' already exists. Not copying from '/etc/skel'.  
adduser: Warning: The home directory '/home/ftpkishore' does not belong to the user you are currently creating.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: password updated successfully  
Changing the user information for ftpkishore  
Enter the new value, or press ENTER for the default  
Full Name []:  
Room Number []:  
Work Phone []:  
Home Phone []:  
Other []:  
Is the information correct? [Y/n] Y
```

Step - 5: Create directories to upload files and set necessary permissions:



```
kishore@kishore-virtual-machine: ~$ sudo mkdir /home/ftpkishore/upload  
kishore@kishore-virtual-machine: ~$ sudo chown ftpkishore:ftpkishore /home/ftpkishore/upload  
kishore@kishore-virtual-machine: ~$ sudo chmod 755 /home/ftpkishore/upload  
kishore@kishore-virtual-machine: ~$
```

Step - 6: Login to FTP Server and move to upload directory in order to upload files:

```
kishore@kishore-virtual-machine:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:kishore): ftpkishore
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
ftp> pwd
Remote directory: /
ftp>
ftp> ls
229 Entering Extended Passive Mode (|||64022|)
150 Here comes the directory listing.
drwxr-xr-x  2 1001    1001      4096 Mar 03 06:47 files
drwxr-xr-x  2 1001    1001      4096 Mar 03 06:33 ftpfiles
drwxr-xr-x  2 1001    1001      4096 Mar 03 07:10 upload
226 Directory send OK.
ftp> cd upload
250 Directory successfully changed.
ftp> lcd
Local directory now: /home/kishore
ftp>
ftp> put test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||27624|)
150 Ok to send data.
100% |*****| 10      6.62 KiB/s   00:00 ETA
226 Transfer complete.
10 bytes sent in 00:00 (4.08 KiB/s)
ftp>
ftp> exit
221 Goodbye.
kishore@kishore-virtual-machine:~$
```

Step - 7: PUT, GET and Delete Files

```
kishore@kishore-virtual-machine:~$ rm test.txt
kishore@kishore-virtual-machine:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:kishore): ftpkishore
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp> exit
221 Goodbye.
kishore@kishore-virtual-machine:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:kishore): ftpkishore
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
ftp> cd upload
250 Directory successfully changed.
ftp>
ftp> get test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||37443|)
150 Opening BINARY mode data connection for test.txt (10 bytes).
100% |*****| 10      21.94 KiB/s   00:00 ETA
226 Transfer complete.
10 bytes received in 00:00 (7.54 KiB/s)
ftp> exit
221 Goodbye.
kishore@kishore-virtual-machine:~$
```

SFTP Server:

Step - 1: Install and Setup SFTP Servers, OpenSSH Server

```
kishore@kishore-virtual-machine:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
kishore@kishore-virtual-machine:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.13).
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
kishore@kishore-virtual-machine:~$
```

Step - 2: Start and Enable SFTP Server:

```
kishore@kishore-virtual-machine: -
kishore@kishore-virtual-machine:~$ sudo systemctl start ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
kishore@kishore-virtual-machine:~$
kishore@kishore-virtual-machine:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-03-03 08:21:20 IST; 2min 40s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 4155 (sshd)
    Tasks: 1 (limit: 4542)
   Memory: 1.9M
      CPU: 59ms
   CGroup: /system.slice/ssh.service
           └─4155 'sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups'

Mar 03 08:21:33 kishore-virtual-machine sshd[4159]: Invalid user sftpuser from 127.0.0.1 port 57906
Mar 03 08:21:38 kishore-virtual-machine sshd[4159]: pam_unix(sshd:auth): check pass; user unknown
Mar 03 08:21:38 kishore-virtual-machine sshd[4159]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Mar 03 08:21:40 kishore-virtual-machine sshd[4159]: Failed password for invalid user sftpuser from 127.0.0.1 port 57906 ssh2
Mar 03 08:21:40 kishore-virtual-machine sshd[4159]: pam_unix(sshd:auth): check pass; user unknown
Mar 03 08:21:40 kishore-virtual-machine sshd[4159]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Mar 03 08:21:50 kishore-virtual-machine sshd[4159]: Failed password for invalid user sftpuser from 127.0.0.1 port 57906 ssh2
Mar 03 08:22:03 kishore-virtual-machine sshd[4159]: pam_unix(sshd:auth): check pass; user unknown
Mar 03 08:22:05 kishore-virtual-machine sshd[4159]: Failed password for invalid user sftpuser from 127.0.0.1 port 57906 ssh2
Mar 03 08:22:06 kishore-virtual-machine sshd[4159]: Connection closed by invalid user sftpuser 127.0.0.1 port 57906 [preauth]
Mar 03 08:22:06 kishore-virtual-machine sshd[4159]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
kishore@kishore-virtual-machine:~$
```

Step - 3: Create a Group for SFTP Users, add a new SFTP User and add the User to the Group

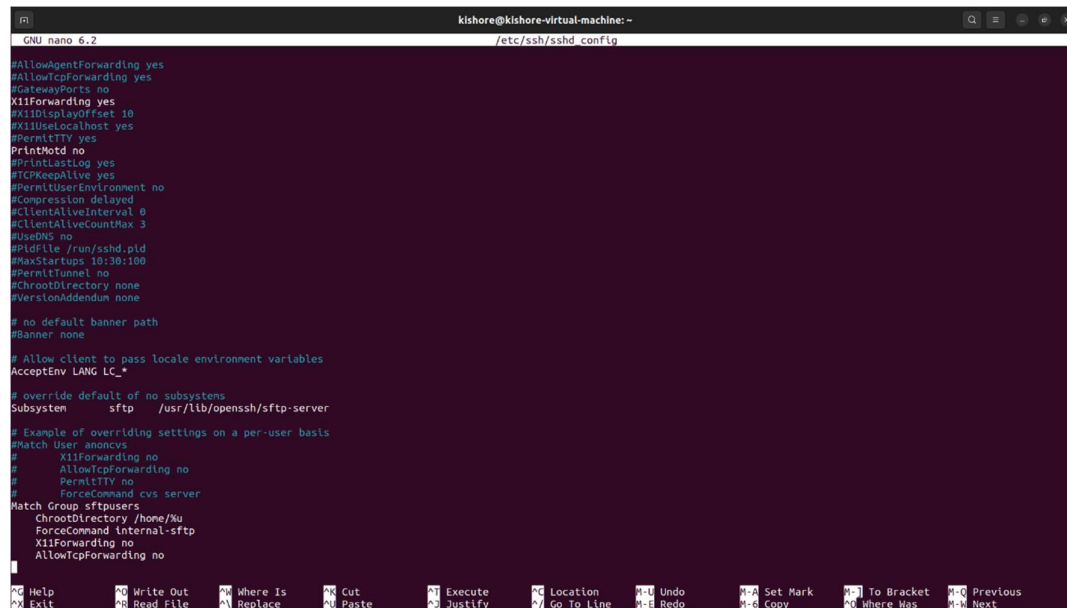
```
kishore@kishore-virtual-machine: -
kishore@kishore-virtual-machine:~$ sudo groupadd sftpkishoregrp
kishore@kishore-virtual-machine:~$
kishore@kishore-virtual-machine:~$ sudo adduser sftpkishore
Adding user 'sftpkishore' ...
Adding new group 'sftpkishore' (1004) ...
Adding new user 'sftpkishore' (1002) with group 'sftpkishore' ...
Creating home directory '/home/sftpkishore' ...
Copying files from '/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for sftpkishore
Enter the new value, or press ENTER for the default
  Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
kishore@kishore-virtual-machine:~$
kishore@kishore-virtual-machine:~$ sudo usermod -aG sftpkishoregrp sftpkishore
kishore@kishore-virtual-machine:~$
```

Step - 4: Create the Upload Directory and Set the Permissions to Upload files

```
kishore@kishore-virtual-machine: -
kishore@kishore-virtual-machine:~$ sudo mkdir -p /home/sftpkishore/upload
kishore@kishore-virtual-machine:~$ sudo chown root:root /home/sftpkishore
kishore@kishore-virtual-machine:~$ sudo chmod 755 /home/sftpkishore
kishore@kishore-virtual-machine:~$ sudo chown sftpkishore:sftpkishore /home/sftpkishore/upload
kishore@kishore-virtual-machine:~$
```

Step - 5: Edit the Config File to set all Necessary Permissions:

`sudo nano /etc/ssh/sshd_config`



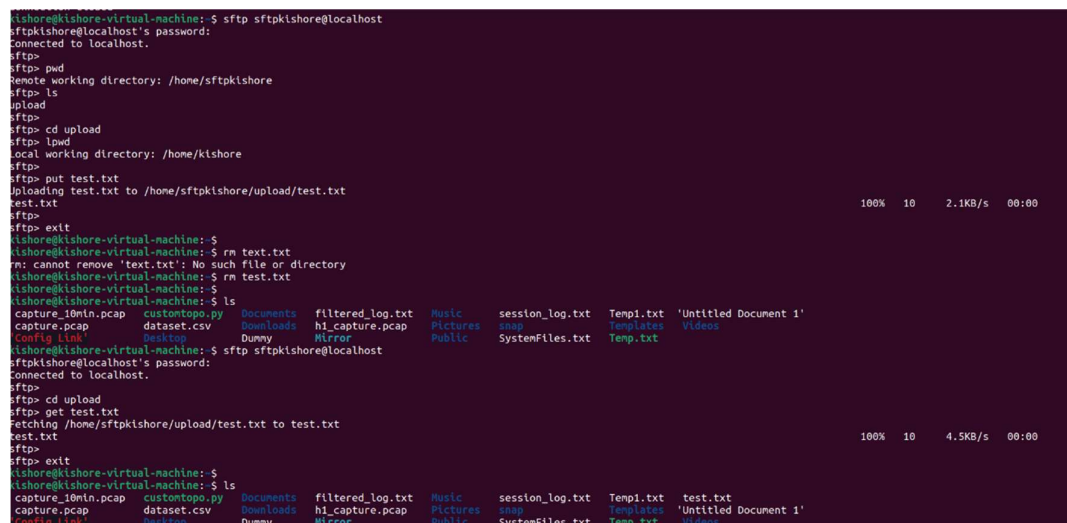
```
kishore@kishore-virtual-machine: ~  
GNU nano 6.2 /etc/ssh/sshd_config  
#AllowAgentForwarding yes  
#AllowTcpForwarding yes  
#GatewayPorts no  
X11Forwarding yes  
#X11DisplayOffset 10  
#X11UseLocalhost yes  
#PermitTTY yes  
PrintMotd no  
#PrintLastLog yes  
#TCPKeepAlive yes  
#PermitUserEnvironment no  
#Compression delayed  
#ClientAliveInterval 0  
#ClientAliveCountMax 3  
#UseDNS no  
#PidFile /run/sshd.pid  
#MaxStartups 10:30:100  
#PermitTunnel no  
#ChrootDirectory none  
#VersionAddendum none  
  
# no default banner path  
#Banner none  
  
# Allow client to pass locale environment variables  
AcceptEnv LANG LC_*  
  
# override default of no subsystems  
Subsystem sftp /usr/lib/openssh/sftp-server  
  
# Example of overriding settings on a per-user basis  
#Match User anoncvs  
#  
#    X11Forwarding no  
#    AllowTcpForwarding no  
#    PermitTTY no  
#    ForceCommand cvs server  
#Match Group sftpushers  
ChrootDirectory /home/kishore  
ForceCommand internal-sftp  
X11Forwarding no  
AllowTcpForwarding no
```

Step - 6: Restart the SFTP Server and Login to the User:



```
kishore@kishore-virtual-machine: ~  
kishore@kishore-virtual-machine:~$ sudo nano /etc/ssh/sshd_config  
kishore@kishore-virtual-machine:~$ sudo systemctl restart ssh  
kishore@kishore-virtual-machine:~$  
kishore@kishore-virtual-machine:~$ sftp sftpuser@localhost  
sftpuser@localhost's password:  
Permission denied, please try again.  
sftpuser@localhost's password:  
Permission denied, please try again.  
sftpuser@localhost's password:  
sftpuser@localhost: Permission denied (publickey,password).  
Connection closed.  
Connection closed.  
kishore@kishore-virtual-machine:~$ sftp sftpkishore@localhost  
sftpkishore@localhost's password:  
Connected to localhost.  
sftp>
```

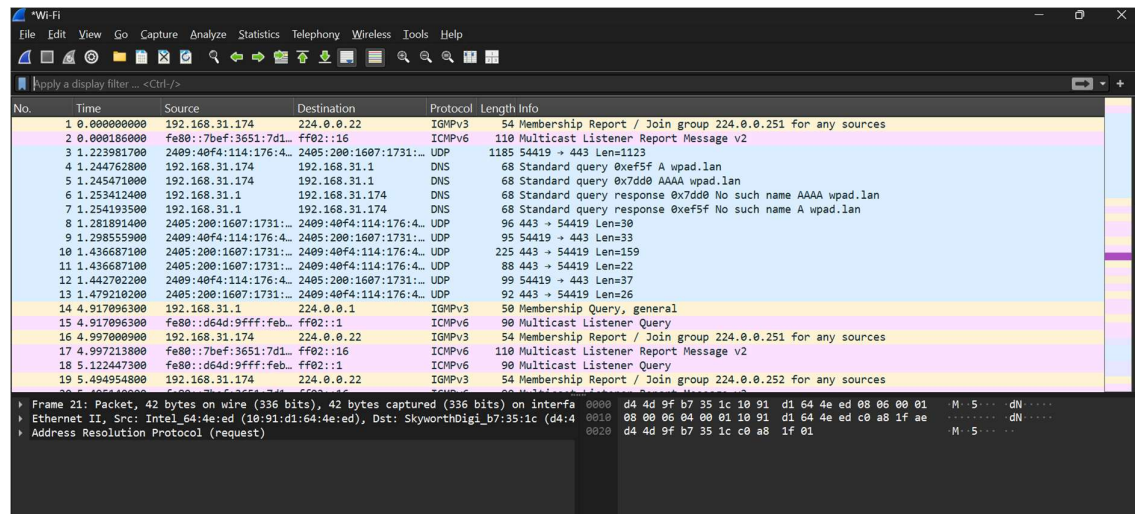
Step - 7: Use the Put method to add file and get method to download the file



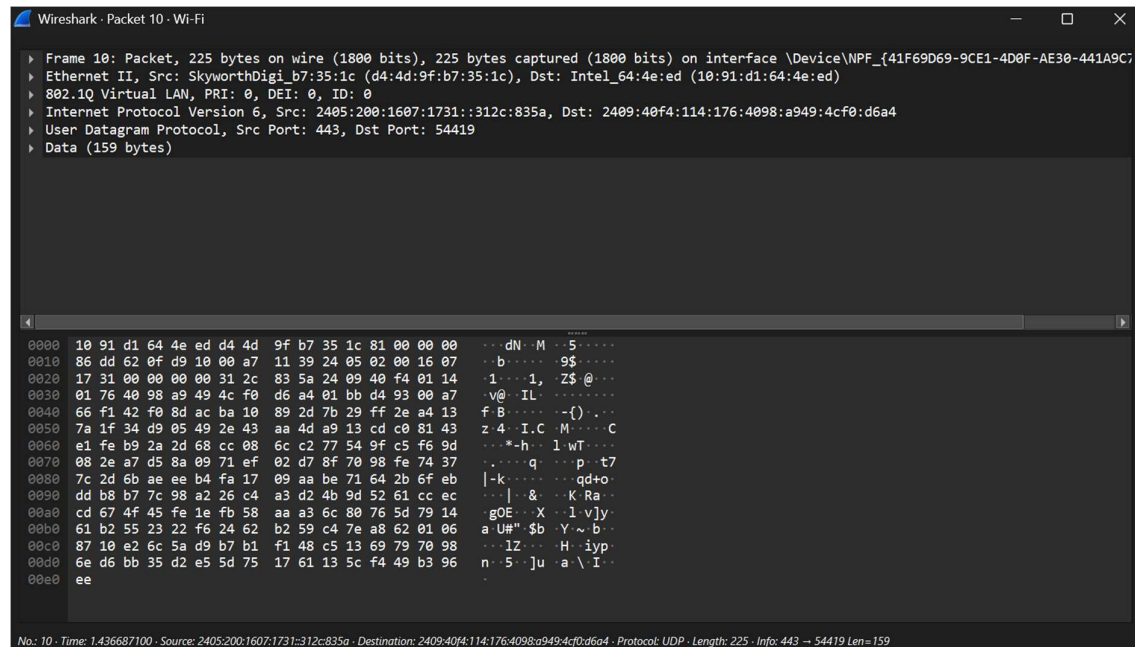
```
kishore@kishore-virtual-machine:~$ sftp sftpkishore@localhost  
sftpkishore@localhost's password:  
Connected to localhost.  
sftp>  
sftp> pwd  
Remote working directory: /home/sftpkishore  
sftp> ls  
upload  
sftp>  
sftp> cd upload  
sftp> lpwd  
Local working directory: /home/kishore  
sftp>  
sftp> put test.txt  
Uploading test.txt to /home/sftpkishore/upload/test.txt  
test.txt 100% 10 2.1KB/s 00:00  
sftp>  
sftp> exit  
kishore@kishore-virtual-machine:~$  
rm: cannot remove 'test.txt': No such file or directory  
kishore@kishore-virtual-machine:~$ rm test.txt  
kishore@kishore-virtual-machine:~$  
kishore@kishore-virtual-machine:~$ sftp sftpkishore@localhost  
sftpkishore@localhost's password:  
Connected to localhost.  
sftp>  
sftp> cd upload  
sftp> get test.txt  
Fetching /home/sftpkishore/upload/test.txt to test.txt  
test.txt 100% 10 4.5KB/s 00:00  
sftp>  
sftp> exit  
kishore@kishore-virtual-machine:~$  
kishore@kishore-virtual-machine:~$ ls  
capture_10min.pcap customtopo.py Documents filtered_log.txt Music session_log.txt Temp1.txt 'Untitled Document 1'  
capture.pcap dataset.csv Downloads hi_capture.pcap Pictures snap Templates Videos  
Config Link Desktop Dummy Mirror Public SystemFiles.txt Temp.txt
```


Question - 3: Explore with Wireshark/TCP-dump/cisco packet tracer tools and learn about packets filters.

Captured Packets:

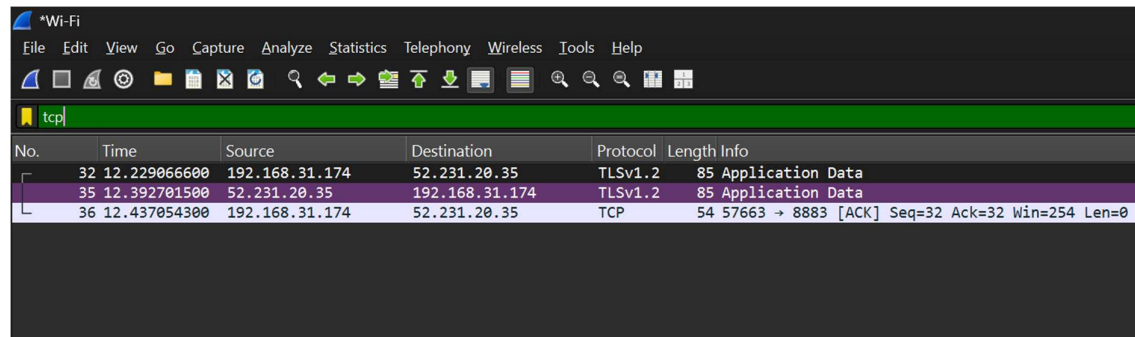


Packet Info:



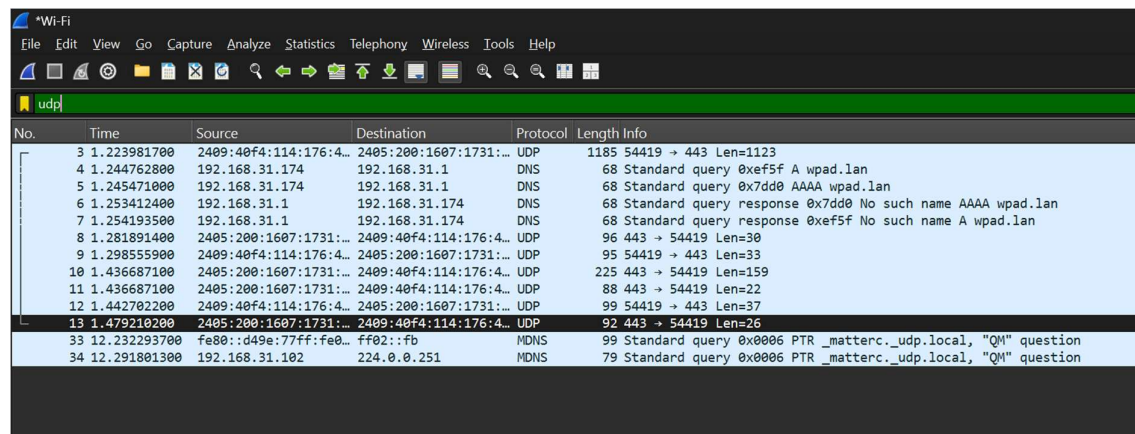
Packet Filters:

TCP:



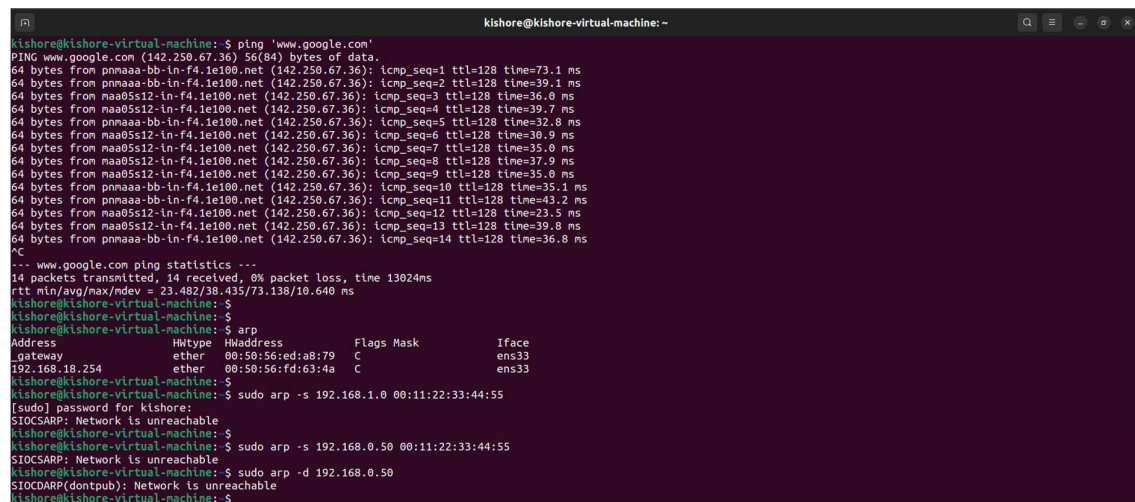
No.	Time	Source	Destination	Protocol	Length	Info
32	12.229066600	192.168.31.174	52.231.20.35	TLSv1.2	85	Application Data
35	12.392701500	52.231.20.35	192.168.31.174	TLSv1.2	85	Application Data
36	12.437054300	192.168.31.174	52.231.20.35	TCP	54	57663 → 8883 [ACK] Seq=32 Ack=32 Win=254 Len=0

UDP



No.	Time	Source	Destination	Protocol	Length	Info
3	1.223981700	2409:40f4:114:176:4...	2405:200:1607:1731:...	UDP	1185	54419 → 443 Len=1123
4	1.244762800	192.168.31.174	192.168.31.1	DNS	68	Standard query 0xef5f A wpad.lan
5	1.245471000	192.168.31.174	192.168.31.1	DNS	68	Standard query 0x7dd0 AAAA wpad.lan
6	1.253412400	192.168.31.1	192.168.31.174	DNS	68	Standard query response 0x7dd0 No such name AAAA wpad.lan
7	1.254193500	192.168.31.1	192.168.31.174	DNS	68	Standard query response 0xef5f No such name A wpad.lan
8	1.281891400	2405:200:1607:1731:...	2409:40f4:114:176:4...	UDP	96	443 → 54419 Len=30
9	1.298555900	2409:40f4:114:176:4...	2405:200:1607:1731:...	UDP	95	54419 → 443 Len=33
10	1.436687100	2405:200:1607:1731:...	2409:40f4:114:176:4...	UDP	225	443 → 54419 Len=159
11	1.436687100	2405:200:1607:1731:...	2409:40f4:114:176:4...	UDP	88	443 → 54419 Len=22
12	1.442702200	2409:40f4:114:176:4...	2405:200:1607:1731:...	UDP	99	54419 → 443 Len=37
13	1.479210200	2405:200:1607:1731:...	2409:40f4:114:176:4...	UDP	92	443 → 54419 Len=26
33	12.232293700	fe80::d49e:77ff:fe0...	ff02::fb	MDNS	99	Standard query 0x0006 PTR _matterc_udp.local, "QM" question
34	12.291801300	192.168.31.102	224.0.0.251	MDNS	79	Standard query 0x0006 PTR _matterc_udp.local, "QM" question

Question – 4: Understand linux utility commands like - ping, arp (Understand each params from ifconfig output)



```
kishore@kishore-virtual-machine: ~  
$ ping 'www.google.com'  
PING www.google.com (142.250.67.36) 56(84) bytes of data:  
64 bytes from pnmaaa-bb-ln-f4.1e100.net (142.250.67.36): icmp_seq=1 ttl=128 time=73.1 ms  
64 bytes from pnmaaa-bb-ln-f4.1e100.net (142.250.67.36): icmp_seq=2 ttl=128 time=39.1 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=3 ttl=128 time=36.0 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=4 ttl=128 time=39.7 ms  
64 bytes from pnmaaa-bb-ln-f4.1e100.net (142.250.67.36): icmp_seq=5 ttl=128 time=32.8 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=6 ttl=128 time=30.9 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=7 ttl=128 time=35.0 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=8 ttl=128 time=37.9 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=9 ttl=128 time=35.0 ms  
64 bytes from pnmaaa-bb-ln-f4.1e100.net (142.250.67.36): icmp_seq=10 ttl=128 time=35.1 ms  
64 bytes from pnmaaa-bb-ln-f4.1e100.net (142.250.67.36): icmp_seq=11 ttl=128 time=43.2 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=12 ttl=128 time=23.5 ms  
64 bytes from maa05s12-ln-f4.1e100.net (142.250.67.36): icmp_seq=13 ttl=128 time=39.8 ms  
64 bytes from pnmaaa-bb-ln-f4.1e100.net (142.250.67.36): icmp_seq=14 ttl=128 time=36.8 ms  
^C  
--- www.google.com ping statistics ---  
14 packets transmitted, 14 received, 0% packet loss, time 13024ms  
rtt min/avg/max/ndev = 23.482/38.435/73.138/10.640 ms  
kishore@kishore-virtual-machine: ~  
$ ifconfig  
Address HWtype HWaddress Flags Mask Iface  
gateway ether 00:50:56:ed:a8:79 C ens33  
192.168.18.254 ether 00:50:56:fd:63:4a C ens33  
kishore@kishore-virtual-machine: ~  
$ sudo arp -s 192.168.1.0 00:11:22:33:44:55  
[sudo] password for kishore:  
SIOCRRARP: Network is unreachable  
kishore@kishore-virtual-machine: ~  
$ sudo arp -s 192.168.0.50 00:11:22:33:44:55  
SIOCRRARP: Network is unreachable  
kishore@kishore-virtual-machine: ~  
$ sudo arp -d 192.168.0.50  
SIOCRRARP(dontpub): Network is unreachable  
kishore@kishore-virtual-machine: ~
```


Question – 5: Understand what happens when duplicate IPs configured in a network.

Using Mininet, Create a simple network with 3 interfaces h1, h2 and h3

```
kishore@kishore-virtual-machine: -
kishore@kishore-virtual-machine:~$ sudo mn --topo single,3 --mac --controller=none
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
s1 ...
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

Configure Interface h2 with the same IP as h1

```
mininet>
mininet> h2 ifconfig h2-eth0 10.0.0.1
mininet>
```

```
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::200:ff:fe00:1 prefixlen 64 scopeid 0x20<link>
    ether 00:00:00:00:00:01 txqueuelen 1000 (Ethernet)
    RX packets 22 bytes 2913 (2.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9 bytes 786 (786.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet>
mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::200:ff:fe00:2 prefixlen 64 scopeid 0x20<link>
    ether 00:00:00:00:00:02 txqueuelen 1000 (Ethernet)
    RX packets 23 bytes 2979 (2.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9 bytes 786 (786.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet>
mininet> h3 ifconfig
h3-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.3 netmask 255.0.0.0 broadcast 10.255.255.255
    inet6 fe80::200:ff:fe00:3 prefixlen 64 scopeid 0x20<link>
    ether 00:00:00:00:00:03 txqueuelen 1000 (Ethernet)
    RX packets 22 bytes 2909 (2.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10 bytes 856 (856.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Tried Ping:

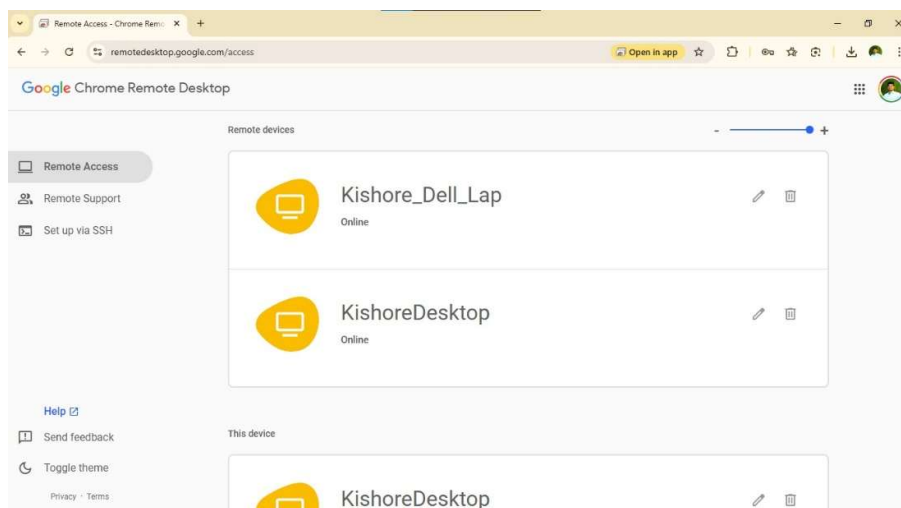
```
kishore@kishore-virtual-machine: ~
$ ip netns exec ns1 ip netns exec ns2 ping -c 1 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
From 10.0.0.3 icmp_seq=1 Destination Host Unreachable
From 10.0.0.3 icmp_seq=2 Destination Host Unreachable
From 10.0.0.3 icmp_seq=3 Destination Host Unreachable
From 10.0.0.3 icmp_seq=4 Destination Host Unreachable
From 10.0.0.3 icmp_seq=5 Destination Host Unreachable
From 10.0.0.3 icmp_seq=6 Destination Host Unreachable
From 10.0.0.3 icmp_seq=7 Destination Host Unreachable
From 10.0.0.3 icmp_seq=8 Destination Host Unreachable
From 10.0.0.3 icmp_seq=9 Destination Host Unreachable
From 10.0.0.3 icmp_seq=10 Destination Host Unreachable
From 10.0.0.3 icmp_seq=11 Destination Host Unreachable
From 10.0.0.3 icmp_seq=12 Destination Host Unreachable
From 10.0.0.3 icmp_seq=13 Destination Host Unreachable
From 10.0.0.3 icmp_seq=14 Destination Host Unreachable
From 10.0.0.3 icmp_seq=15 Destination Host Unreachable
From 10.0.0.3 icmp_seq=16 Destination Host Unreachable
From 10.0.0.3 icmp_seq=17 Destination Host Unreachable
From 10.0.0.3 icmp_seq=18 Destination Host Unreachable
From 10.0.0.3 icmp_seq=19 Destination Host Unreachable
From 10.0.0.3 icmp_seq=20 Destination Host Unreachable
From 10.0.0.3 icmp_seq=21 Destination Host Unreachable
From 10.0.0.3 icmp_seq=22 Destination Host Unreachable
From 10.0.0.3 icmp_seq=23 Destination Host Unreachable
From 10.0.0.3 icmp_seq=24 Destination Host Unreachable
^C
--- 10.0.0.1 ping statistics ---
26 packets transmitted, 0 received, +24 errors, 100% packet loss, time 25619ms
pipe 4
mininet> mininet> mininet> mininet> h3 ping 10.0.0.1
mininet> h3 arp -n
Address HWtype Hwaddress Flags Mask Iface
10.0.0.1 HWtype Hwaddress (incomplete) Flags Mask Iface
mininet> h3 arp -n
Address HWtype Hwaddress Flags Mask Iface
10.0.0.1 HWtype Hwaddress (incomplete) Flags Mask Iface
mininet>
```

Packets getting lost due to duplicate IPs. Hence, if we configure Duplicate IPs, then:

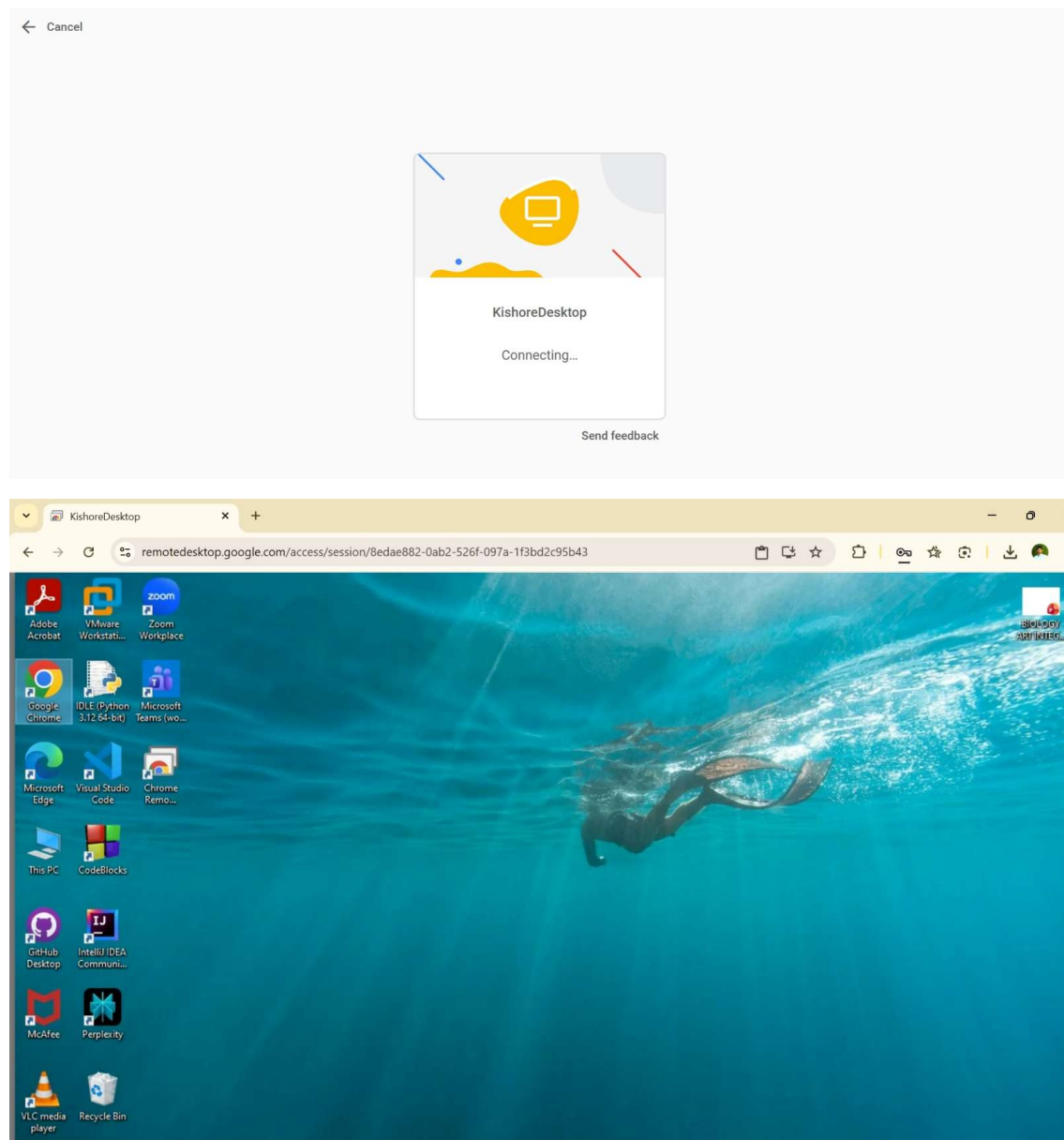
- ❖ Packets may get lost
- ❖ ARP table may keep changing and hence P keeps getting mapped to different MAC addresses.
- ❖ In some cases, even the OS might warn of duplicate IPs.

Question – 6: Understand how to access remote system using (VNC viewer, Anydesk, team viewer and remote desktop connections)

View from Desktop:



Accessing Desktop from Laptop:



Question - 7. How to check your default gateway is reachable or not and understand about default gateway.

Step - 1: Identify Default Gateway

command used: ip route

```
kishore@kishore-virtual-machine: -
kishore@kishore-virtual-machine: $
kishore@kishore-virtual-machine: $
kishore@kishore-virtual-machine: $ ip route
default via 192.168.18.2 dev ens33 proto dhcp metric 100
169.254.0.0/16 dev ens33 scope link metric 1000
192.168.18.0/24 dev ens33 proto kernel scope link src 192.168.18.128 metric 100
kishore@kishore-virtual-machine: $
```

Step - 2: Ping the Identified Default Gateway Identified in order to See if the Interface is active:

```
kishore@kishore-virtual-machine: -
kishore@kishore-virtual-machine: $ ping 192.168.18.2
PING 192.168.18.2 (192.168.18.2) 56(84) bytes of data:
64 bytes from 192.168.18.2: icmp_seq=1 ttl=128 time=1.19 ms
64 bytes from 192.168.18.2: icmp_seq=2 ttl=128 time=0.448 ms
64 bytes from 192.168.18.2: icmp_seq=3 ttl=128 time=0.927 ms
64 bytes from 192.168.18.2: icmp_seq=4 ttl=128 time=0.868 ms
64 bytes from 192.168.18.2: icmp_seq=5 ttl=128 time=0.868 ms
64 bytes from 192.168.18.2: icmp_seq=6 ttl=128 time=0.441 ms
64 bytes from 192.168.18.2: icmp_seq=7 ttl=128 time=0.593 ms
64 bytes from 192.168.18.2: icmp_seq=8 ttl=128 time=0.444 ms
64 bytes from 192.168.18.2: icmp_seq=9 ttl=128 time=0.750 ms
64 bytes from 192.168.18.2: icmp_seq=10 ttl=128 time=0.761 ms
64 bytes from 192.168.18.2: icmp_seq=11 ttl=128 time=0.643 ms
64 bytes from 192.168.18.2: icmp_seq=12 ttl=128 time=0.502 ms
64 bytes from 192.168.18.2: icmp_seq=13 ttl=128 time=0.530 ms
64 bytes from 192.168.18.2: icmp_seq=14 ttl=128 time=0.330 ms
64 bytes from 192.168.18.2: icmp_seq=15 ttl=128 time=0.375 ms
64 bytes from 192.168.18.2: icmp_seq=16 ttl=128 time=0.320 ms
64 bytes from 192.168.18.2: icmp_seq=17 ttl=128 time=0.373 ms
64 bytes from 192.168.18.2: icmp_seq=18 ttl=128 time=0.759 ms
64 bytes from 192.168.18.2: icmp_seq=19 ttl=128 time=0.306 ms
64 bytes from 192.168.18.2: icmp_seq=20 ttl=128 time=0.630 ms
64 bytes from 192.168.18.2: icmp_seq=21 ttl=128 time=1.159 ms
64 bytes from 192.168.18.2: icmp_seq=22 ttl=128 time=0.609 ms
64 bytes from 192.168.18.2: icmp_seq=23 ttl=128 time=0.932 ms
64 bytes from 192.168.18.2: icmp_seq=24 ttl=128 time=0.882 ms
64 bytes from 192.168.18.2: icmp_seq=25 ttl=128 time=0.846 ms
64 bytes from 192.168.18.2: icmp_seq=26 ttl=128 time=0.587 ms
64 bytes from 192.168.18.2: icmp_seq=27 ttl=128 time=0.297 ms
64 bytes from 192.168.18.2: icmp_seq=28 ttl=128 time=0.348 ms
64 bytes from 192.168.18.2: icmp_seq=29 ttl=128 time=0.588 ms
64 bytes from 192.168.18.2: icmp_seq=30 ttl=128 time=0.268 ms
64 bytes from 192.168.18.2: icmp_seq=31 ttl=128 time=0.320 ms
64 bytes from 192.168.18.2: icmp_seq=32 ttl=128 time=0.538 ms
64 bytes from 192.168.18.2: icmp_seq=33 ttl=128 time=0.329 ms
64 bytes from 192.168.18.2: icmp_seq=34 ttl=128 time=0.708 ms
64 bytes from 192.168.18.2: icmp_seq=35 ttl=128 time=0.847 ms
64 bytes from 192.168.18.2: icmp_seq=36 ttl=128 time=0.517 ms
64 bytes from 192.168.18.2: icmp_seq=37 ttl=128 time=0.272 ms
64 bytes from 192.168.18.2: icmp_seq=38 ttl=128 time=0.603 ms
64 bytes from 192.168.18.2: icmp_seq=39 ttl=128 time=0.473 ms
64 bytes from 192.168.18.2: icmp_seq=40 ttl=128 time=0.881 ms
64 bytes from 192.168.18.2: icmp_seq=41 ttl=128 time=0.459 ms
64 bytes from 192.168.18.2: icmp_seq=42 ttl=128 time=0.608 ms
64 bytes from 192.168.18.2: icmp_seq=43 ttl=128 time=1.27 ms
64 bytes from 192.168.18.2: icmp_seq=44 ttl=128 time=0.468 ms
64 bytes from 192.168.18.2: icmp_seq=45 ttl=128 time=0.339 ms
```

```
kishore@kishore-virtual-machine: ~  
64 bytes from 192.168.18.2: icmp_seq=57 ttl=128 time=0.917 ms  
64 bytes from 192.168.18.2: icmp_seq=58 ttl=128 time=0.483 ms  
64 bytes from 192.168.18.2: icmp_seq=59 ttl=128 time=0.372 ms  
64 bytes from 192.168.18.2: icmp_seq=60 ttl=128 time=0.406 ms  
64 bytes from 192.168.18.2: icmp_seq=61 ttl=128 time=0.470 ms  
64 bytes from 192.168.18.2: icmp_seq=62 ttl=128 time=1.77 ms  
64 bytes from 192.168.18.2: icmp_seq=63 ttl=128 time=1.06 ms  
64 bytes from 192.168.18.2: icmp_seq=64 ttl=128 time=0.547 ms  
64 bytes from 192.168.18.2: icmp_seq=65 ttl=128 time=1.03 ms  
64 bytes from 192.168.18.2: icmp_seq=66 ttl=128 time=1.02 ms  
64 bytes from 192.168.18.2: icmp_seq=67 ttl=128 time=1.08 ms  
64 bytes from 192.168.18.2: icmp_seq=68 ttl=128 time=0.190 ms  
64 bytes from 192.168.18.2: icmp_seq=69 ttl=128 time=0.359 ms  
64 bytes from 192.168.18.2: icmp_seq=70 ttl=128 time=0.859 ms  
64 bytes from 192.168.18.2: icmp_seq=71 ttl=128 time=1.02 ms  
64 bytes from 192.168.18.2: icmp_seq=72 ttl=128 time=0.912 ms  
64 bytes from 192.168.18.2: icmp_seq=73 ttl=128 time=0.483 ms  
64 bytes from 192.168.18.2: icmp_seq=74 ttl=128 time=0.299 ms  
64 bytes from 192.168.18.2: icmp_seq=75 ttl=128 time=0.346 ms  
64 bytes from 192.168.18.2: icmp_seq=76 ttl=128 time=0.333 ms  
64 bytes from 192.168.18.2: icmp_seq=77 ttl=128 time=0.451 ms  
64 bytes from 192.168.18.2: icmp_seq=78 ttl=128 time=0.744 ms  
64 bytes from 192.168.18.2: icmp_seq=79 ttl=128 time=0.367 ms  
64 bytes from 192.168.18.2: icmp_seq=80 ttl=128 time=0.305 ms  
64 bytes from 192.168.18.2: icmp_seq=81 ttl=128 time=0.403 ms  
64 bytes from 192.168.18.2: icmp_seq=82 ttl=128 time=1.51 ms  
64 bytes from 192.168.18.2: icmp_seq=83 ttl=128 time=0.594 ms  
64 bytes from 192.168.18.2: icmp_seq=84 ttl=128 time=0.730 ms  
64 bytes from 192.168.18.2: icmp_seq=85 ttl=128 time=0.850 ms  
64 bytes from 192.168.18.2: icmp_seq=86 ttl=128 time=0.796 ms  
64 bytes from 192.168.18.2: icmp_seq=87 ttl=128 time=0.302 ms  
64 bytes from 192.168.18.2: icmp_seq=88 ttl=128 time=0.529 ms  
64 bytes from 192.168.18.2: icmp_seq=89 ttl=128 time=0.627 ms  
64 bytes from 192.168.18.2: icmp_seq=90 ttl=128 time=0.689 ms  
64 bytes from 192.168.18.2: icmp_seq=91 ttl=128 time=0.654 ms  
64 bytes from 192.168.18.2: icmp_seq=92 ttl=128 time=0.457 ms  
64 bytes from 192.168.18.2: icmp_seq=93 ttl=128 time=0.504 ms  
64 bytes from 192.168.18.2: icmp_seq=94 ttl=128 time=0.886 ms  
64 bytes from 192.168.18.2: icmp_seq=95 ttl=128 time=0.688 ms  
64 bytes from 192.168.18.2: icmp_seq=96 ttl=128 time=0.562 ms  
64 bytes from 192.168.18.2: icmp_seq=97 ttl=128 time=0.527 ms  
64 bytes from 192.168.18.2: icmp_seq=98 ttl=128 time=0.230 ms  
^C  
--- 192.168.18.2 ping statistics ---  
98 packets transmitted, 98 received, 0% packet loss, time 98522ms  
rtt min/avg/max/mdev = 0.190/0.643/1.771/0.317 ms  
kishore@kishore-virtual-machine: $
```

0 % packet loss. Hence, the default Gateway is active.

Question - 8. Check iwconfig/ifconfig to understand in detail about network interfaces (check about interface speed, MTU and other parameters)

```
kishore@kishore-virtual-machine: ~  
$ ifconfig  
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.18.128 netmask 255.255.255.0 broadcast 192.168.18.255  
    inet6 fe80::8b13:3954:e3d4:510 prefixlen 64 scopeid 0x20<link>  
    ether 08:0c:29:2b:fc:2a txqueuelen 1000 (Ethernet)  
    RX packets 31021 bytes 27804347 (27.0 MB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 18469 bytes 9450443 (9.4 MB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 1270 bytes 126790 (126.7 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1270 bytes 126790 (126.7 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
kishore@kishore-virtual-machine: $
```

ifconfig command Output Parameters:

- ens33 → Main network adapter (VM Ethernet)
- lo → Loopback interface (internal communication)

Status Flags

- UP → Interface is enabled
- RUNNING → Network link is active
- BROADCAST → Can send broadcast packets
- MULTICAST → Supports multicast
- mtu 1500 => Maximum packet size = 1500 bytes (standard Ethernet size).
- inet 192.168.18.128
netmask 255.255.255.0
broadcast 192.168.18.255

Hence, the Ip Address of my Device is:

- 192.168.18.128
- and Subnet Mask Is:
- 255.255.255.0 → /24 network

Other Parameters:

- IPV6 Address:
- inet6 fe80::8b13:3954:e3d4:510
- MAC Address:
- ether 00:0c:29:2b:fc:2a
- Loop Back Interface (Localhost IP):
- 127.0.0.1

RX (Received)

RX packets 31021

bytes 27804347 (27.8 MB)

errors 0

dropped 0

- Received 31,021 packets
- Total 27.8 MB data
- No errors
- No packet drops

TX (Transmitted)

TX packets 18469

bytes 9450443 (9.4 MB)

errors 0

dropped 0

collisions 0

- Sent 18,469 packets
- 9.4 MB transmitted
- No transmission errors
- No collisions

Question – 9: Log in to your home router's web interface (usually at 192.168.1.1 or 192.168.0.1) and check the connected devices list.



Question – 10: Explain how a DHCP server assigns IP addresses to devices in your network.

The client first enters the DHCP State and then, the following DORA (Discover, Offer, Request and Accept) Process takes place:

- Client does a general broadcast and sends a discover message
- The DHCP server then does a unicast to the client and offers messages with the allocated IP, Subnet mask, DNS etc., in a packet
- Then, the DHCP client Accepts the IP address and responds with a Request Data message.
- DHCP server then sends an ack to confirm that the IP and MAC addresses are used and updates its ARP table with the IP -> MAC Mapping.

Question – 11: Using a terminal, connect to a remote machine via SSH and telnet.

Host Machine (Wnidows PC) Source:

```
kishore@kishore-virtual-mac X + v
C:\Users\kishore>ssh kishore@192.168.31.226
The authenticity of host '192.168.31.226 (192.168.31.226)' can't be established.
ED25519 key fingerprint is SHA256:9u3kZK5jBS0Nc0eQZpNj1yq/ui/xZLMJ3DiTsAvRIDg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.31.226' (ED25519) to the list of known hosts.
kishore@192.168.31.226's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

42 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

kishore@kishore-virtual-machine:~$
```

Target Machine (VM - Linux):

```
kishore@kishore-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:0c:29:2b:fc:2a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.31.226/24 brd 192.168.31.255 scope global dynamic noprefroute ens33
        valid_lft 28707sec preferred_lft 28707sec
    inet6 2409:a0f4:114:176:3c6c:c75:532f:10ba/64 scope global dynamic mngtaddr noprefroute
        valid_lft 10808sec preferred_lft 10808sec
    inet6 fe80::8b13:3954:e3d4:510/64 scope link noprefroute
        valid_lft forever preferred_lft forever
kishore@kishore-virtual-machine:~$ sudo systemctl start ssh
[sudo] password for kishore:
kishore@kishore-virtual-machine:~$ sudo systemctl start ssh
kishore@kishore-virtual-machine:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
kishore@kishore-virtual-machine:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2024-03-03 14:02:06 IST; 2min 26s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1125 (sshd)
     Tasks: 1 (limit: 4542)
    Memory: 2.7M
       CPU: 4ms
   CGroup: /system.slice/ssh.service
           └─1125 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 03 14:02:06 kishore-virtual-machine systemd[1]: Starting OpenBSD Secure Shell server...
Mar 03 14:02:06 kishore-virtual-machine sshd[1125]: Server listening on 0.0.0.0 port 22.
Mar 03 14:02:06 kishore-virtual-machine sshd[1125]: Server listening on *: port 22.
Mar 03 14:02:06 kishore-virtual-machine systemd[1]: Started OpenBSD Secure Shell server.
kishore@kishore-virtual-machine:~$
```